FANUC Teach Pendant — Quick Reference

EE0849 Introduction to Robotics | Basri Erdogan

R-30iB Teach Pendant — Front

Teach Pendant — Back (deadman switches)

⚠ Deadman Switch

The yellow switches on the back are **3-position**:

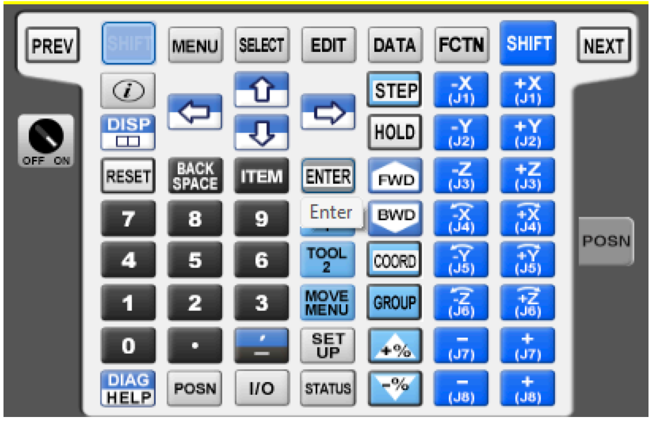
Released	→	robot disabled
Half-press	→	**robot enabled**
Full squeeze	→	emergency stop

E-Stop: Red mushroom button at top-right of pendant. Push to stop. Twist clockwise to release.

Operating Modes

T1	Teach — slow ($\lesssim$ 250 mm/s), deadman req.
T2	Teach — full speed, deadman req.
AUTO	Run — safety gate must be closed

Key Buttons & Commands



Keypad layout (virtual teach pendant)

Button	Function
SHIFT	Hold together with jog keys to move
COORD	Cycle: JOINT / WORLD / TOOL / USER
FWD / BWD	Step through program lines
STEP	Toggle single-step mode
HOLD	Pause running program
MENU	Open top menu
SELECT	Choose / create programs
POSN	Show current position
+% / -%	Speed override up / down
RESET	Clear faults (after E-Stop release)

Motion Commands

`J P[n] % FINE`	Joint move — curved path, fast
`L P[n] mm/s FINE`	Linear move — straight path
`C P[n] ...`	Circular arc through points

Termination types:

`FINE` = stop at point
`CNT`*n* = blend through (0 = tight, 100 = wide)

Teaching a Position

1. Jog robot to desired pose
2. **MENU** → POSITION → Position Register
3. Select slot (P[1], P[2], …)
4. Press **SHIFT + F5 [RECORD]**

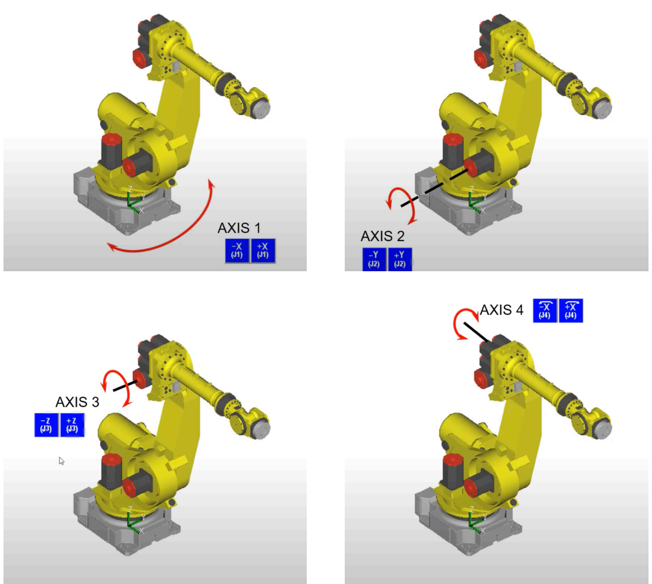
Positions store joint angles (J1–J6) or Cartesian pose (X, Y, Z, W, P, R) plus configuration flags.

Coordinate Frames & Jogging

Joint Mode

Joint Jogging

Each key (J1–J6) moves **one motor** independently. Path is curved — good for fast repositioning.

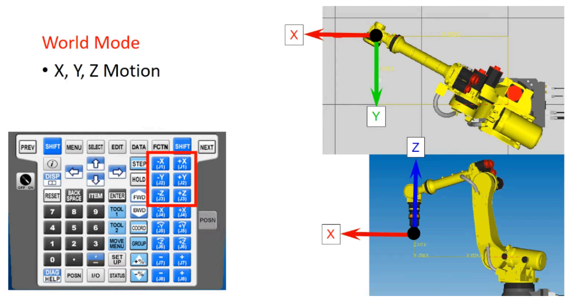


Axes 1–4: red arrows show $\pm$ rotation

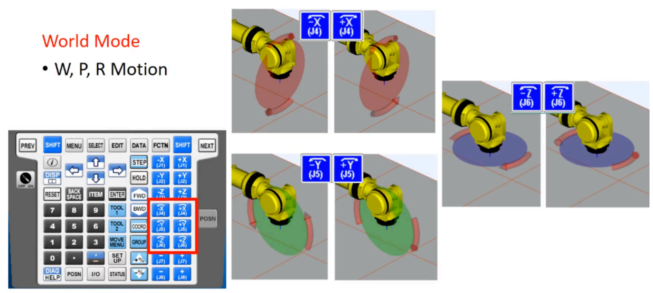
World Mode

World Jogging

End-effector moves in **straight lines** along World frame axes (X/Y/Z + W/P/R).

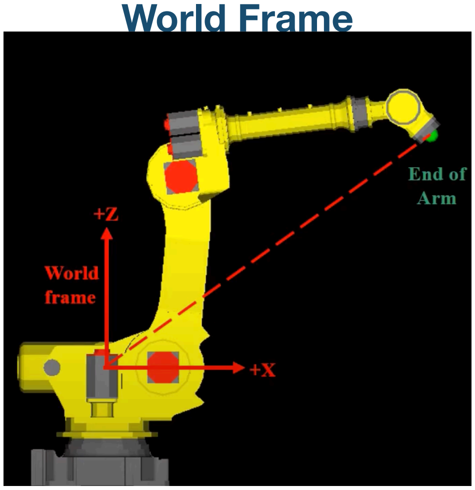


X, Y, Z — linear translation



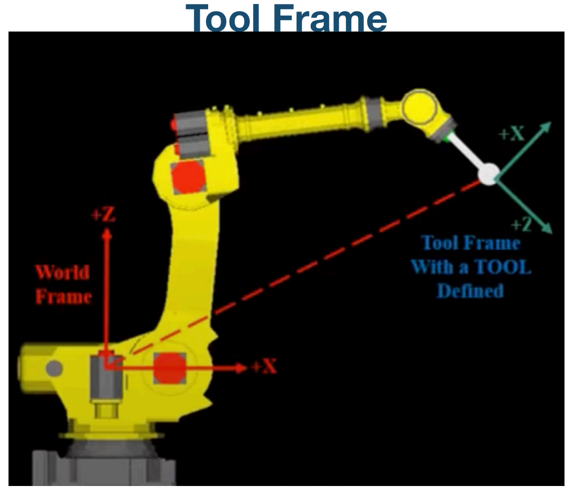
W (Yaw), P (Pitch), R (Roll) — rotation

World Frame



Origin at robot base. $+Z$ up, $+X$ forward.

Tool Frame



Tool Frame with a tool defined — TCP at tool tip.

Frame	Origin	When to Use
JOINT	Each motor independently	Fast repositioning, clearing obstacles
WORLD	Robot base (fixed)	Straight-line paths, teaching positions
TOOL	End-effector / TCP	Precise tool-relative adjustments
USER	Custom (user-defined)	Align jog axes with workpiece or fixture